

**Instructions:**

1. Fill up the necessary informations in the answer script properly.
2. Answer any 3 from the **first 4 experiments only** that you have done. Answering from other experiments will be considered as cheating.

Name: **MOHAMMAD HABIBUL HASSAN KHAN**ID: **22321072**Theory section: **17**Group: **1****E2: End Correction of Meter Bridge**

- What is end correction? On which principle does the meter bridge operate? Why do we use a galvanometer?

E3: Specific Resistance

- What is specific resistance? What are the differences between resistance and specific resistance? On which properties does the specific resistance depend?

E4: Resistance of Galvanometer

- What is the resistance of a galvanometer? By which method we can determine the resistance of a galvanometer? What is the use of shunt in this experiment?

E5: Time Constant of RC circuit

- What is time constant in an RC circuit? Write down the charging and discharging equations of an RC circuit.

E6: Comparing EMF of two cells

- What is electromotive force of a cell? Derive potential divider rule between two resistors connected in series.

E7: Internal Resistance of a cell

- What is internal resistance of a cell? What are the ideal characteristics of a potentiometer?

E8: Mechanical Equivalent of heat

- Define Mechanical Equivalent of heat, Specific heat and Calorie. What is radiation correction?

E9: Unknown Frequency, Time Base of Oscilloscope

- When a Lissajous curve is formed from two sinusoidal signals? Define frequency and Amplitude. In a Lissajous figure, there are three loops. If the smaller frequency is 33 hz, what is the larger frequency?

$$\frac{P}{2} = \frac{R}{3} = \frac{l}{100-l}$$

$$r = \frac{P}{2} = \frac{l+x}{100-l+x} \text{ m}$$

$$x = \frac{l-rL}{r-1}, \quad y = \frac{rL-l}{r-1} - 100$$

E2: End correction of a meter bridge is the correction need to be made to a meter wire.

ideally we assume:

- ① wire is uniform
- ② 100 cm exactly
- ③ the 0 & 100 coincide perfectly with ends of wire
- ④ the metal plate is of 0 resistance that connects the wire.

~~in reality~~

~~in reality:~~ However, in practice this is not true.

∴ End corrections need to be made for accurate calculations

→ we use galvanometer to determine the specific condition when no current passes through it as the equilibrium condition. or the null point.

E5: $\tau = RC$. Time constant for charging process is the amount

→ of time needed to make the potential difference within the capacitors plates 0.63% of E from fully discharged condition

→ time constant for discharging ckt is the amount of time needed to discharge enough charge to make the potential difference of the capacitor ~~0.37%~~ $0.37E$ from initial condition.

→ The time constant indicates the rate of growth or decay of the charging/discharging process.

$$\rightarrow V_c(t) = E(1 - e^{-t/RC}) \quad \leftarrow \text{charging}$$

$$\rightarrow V_c(t) = E e^{-t/RC}$$

Eq:

* Resistance of galvanometer is the resistance of the coil which is wrapped around a frame inside the magnet.

* we can determine the resistance by using half deflection method.

* the use of shunt is to allow a small current to flow which is used to compare ~~with~~

→ For the shunt to be relatively small allows for it to be omitted from the equation giving $R_g \approx G$

→ with smaller S value, the approximation is closer to the exact value of R_g .

~~Eq: properties of~~

$$G \approx \frac{R_1}{\left(\frac{Q}{G} - 1\right)}$$

$$Q = R_1 = 0$$

$$Q = R_1 \neq 0$$

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→ Half deflection method is only valid for high resistance galvanometer.

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